

Lab 4 - VibeCoding

Name: B V TEJAS

SRN: PES1UG24CS108

Section: B

Assigned project: 41_fruit-ninja (Pygame Fruit Ninja)

1. Repository links

Lab repository, Lab-4 folder: <https://github.com/Tejas6499/SE-Labs--PES1UG24CS108/tree/main/Lab-4>

Updated code with one commit per task: https://github.com/Tejas6499/41_fruit-ninja

Assigned repository: https://github.com/SETAPESU26/41_fruit-ninja

2. Tasks completed

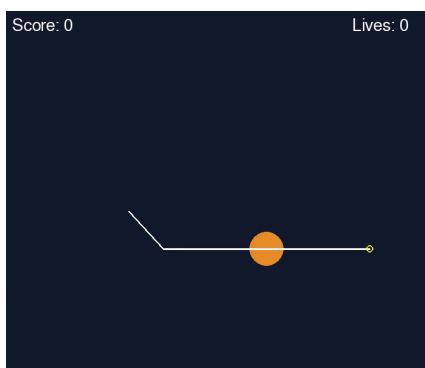
Task	Change made	Commit
1	Slice detection now tests the line segment between the previous and current mouse positions against the fruit, so fast swipes register. The stroke is reset when the pointer leaves the window.	48e6f6d Task 1: fix slice detection for fast swipes
2	A game-over screen dims the playfield and shows GAME OVER and the final score. The console print is removed and slicing stops once the game is over.	a81c104 Task 2: add on-screen game-over screen
3	From the game-over screen, E, M or H starts a new game on Easy, Medium or Hard and Q quits. Each difficulty sets the spawn interval, bomb chance and launch speed. Restart resets all game state.	fa34f56 Task 3: add replay with Easy/Medium/Hard and quit
4	Sound effects for slicing a fruit, slicing a bomb and game over, generated in code with no audio files. The game runs silently if no audio device is available.	ffbceba Task 4: add generated sound effects

Files changed: game/fruit.py (Task 1) and game/game_engine.py (Tasks 1 to 4). main.py and requirements.txt are unchanged.

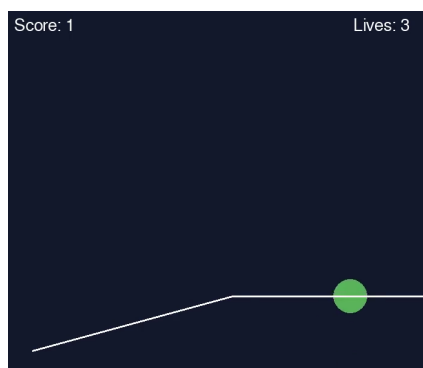
3. Before and after videos

Before: <https://github.com/Tejas6499/SE-Labs--PES1UG24CS108/blob/main/Lab-4/before.mp4>

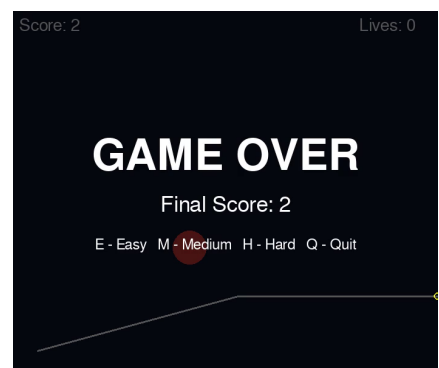
After: <https://github.com/Tejas6499/SE-Labs--PES1UG24CS108/blob/main/Lab-4/after.mp4>



Before: the swipe crosses the fruit, score stays 0 and no game-over screen appears



After: a fast swipe slices the fruit and the score increases



After: game-over screen with final score and replay options

4. Prompts used

All four prompts were given in one chat. Full chat history:

<https://claude.ai/share/de226c97-b7b7-4bd0-80b1-868cdf862069>

Prompt 1 - Task 1 (collision detection)

have a Pygame Fruit Ninja game (files below). In `game/fruit.py`, `contains_point(x, y)` only tests whether a single mouse position lies inside the fruit's circle, and `_handle_motion` in `game/game_engine.py` calls it once per `MOUSEMOTION` event. A fast swipe jumps from outside the fruit to past it between two consecutive events, so the fruit is never sliced even though the blade visually crossed it.

Fix it by testing the line segment between the previous and current mouse positions against the fruit's circle. Add an `intersects_segment(x1, y1, x2, y2)` method to `Fruit` using point-to-segment distance, and have `_handle_motion` track the last position and use it. Return the complete updated `fruit.py` and `game_engine.py`.

Prompt 2 - Task 2 (game-over screen)

Task 2: add a proper game-over screen. Work from the `fruit.py` and `game_engine.py` you just gave me.

Right now, when the game ends, `render()` only prints "Game over!" to the console. Replace that with a real on-screen game-over screen:

- Dim the playfield with a semi-transparent dark overlay.
- Show "GAME OVER" in large text, centred.
- Show "Final Score: N" below it, centred.
- Remove the console print and the `_game_over_logged` flag.
- Once `game_over` is True, stop slicing: `_handle_motion` must not slice fruit or change the score or lives. It's fine to keep drawing the trail.

Keep the swept-segment slice detection working. Don't add replay or sound yet. Return the complete updated `game_engine.py`, and tell me if any other file needs to change.

Prompt 3 - Task 3 (replay with difficulty, or exit)

Task 3: add replay with difficulty selection, or exit. Work from the `game_engine.py` you just gave me.

On the game-over screen:

- Show the options below the final score:
"E - Easy M - Medium H - Hard Q - Quit".
- Pressing E, M or H starts a new game on that difficulty. Pressing Q closes the game cleanly.
- Each difficulty sets `spawn_interval`, `bomb_chance` and `speed_scale`. Medium must match the current values (55, 0.15, 1.0); Easy is slower with fewer bombs, Hard is faster with more bombs.
- Restart must fully reset score, lives, fruits, trail, the spawn timer, `_last_pos` and `game_over`.
- The keys must do nothing while a game is in progress.
- The first game still starts straight away on Medium, as it does now.

Keep the swept-segment slicing and the game-over overlay working. Don't add sound yet. Return the complete updated `game_engine.py`, and `main.py` too if it needs to change for quitting.

Prompt 4 - Task 4 (sound effects)

Task 4: add sound effects. Work from the `game_engine.py` you just gave me.

- Play a short sound when a fruit is sliced, a different one when a bomb is sliced, and a game-over sound when the game ends (either by bomb or by running out of lives). The game-over sound must play once per game, not every frame.
- Generate the sounds in code (for example short tones or noise bursts built with Python's `array` module or `pygame.sndarray`), so no audio files have to be added to the repo.
- If the mixer can't be initialised (no audio device), the game must still run normally with sound disabled, not crash.
- Don't play any sound for swipes after game over.

Keep slicing, the game-over screen and replay/quit working exactly as they are. Return the complete updated `game_engine.py`, and `requirements.txt` and `main.py` too if either needs to change.

5. Chat history

Chat link with the full history: <https://claude.ai/share/de226c97-b7b7-4bd0-80b1-868cdf862069>

A PDF copy of the chat is in the Lab-4 folder as `chat_history.pdf`.